

DMMST: Experimental Results

Dynamic Multi-modal Multi-event Survival Transformer

Results of notebooks 01–11 (October 1, 2026)

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Note on SUPPORT (found after these runs). The numeric preprocessing used METABRIC’s column layout for every dataset. On SUPPORT, five physiological measurements entered the models unscaled and mean blood pressure was reduced to a constant. This affects every SUPPORT number on the numeric path in Tables 1 and 5 (all models alike, including Cox). The parser is fixed and the SUPPORT experiments are being re-run (notebook 12); METABRIC and all other datasets are unaffected.

Summary of what we know

- **The base model is strong.** The transformer with the piecewise-constant-hazard likelihood ($\mathcal{L}_{\text{PCH}}$) is the best or tied-best model among those we ran on METABRIC, SUPPORT, hsa_synthetic, EBMT and the simulator, with the best calibration among neural models. On METABRIC it matches the published PC-Hazard and Cox values; it remains below the published SurvTRACE (0.675 vs. 0.691 on METABRIC, 0.608 vs. 0.640 on SUPPORT).

- **It orders events within a patient best among learned models**, measured against the simulator’s true event times. DeepHit (a competing-risk model) ranks patients well but orders events within a patient badly, which supports the paper’s general multi-event formulation.
- **The penalty terms do not help.** $\mathcal{L}_{\text{rank}}$ and $\mathcal{L}_{\text{mul}}$ never improve discrimination, calibration or true event ordering; tuned, they become neutral; with the original sharpness ($\sigma = 0.1$) they hurt.
- **The within-subject C-index can be gamed.** $\mathcal{L}_{\text{mul}}$ raises it above the oracle while making the true ordering worse. Proper scores against the true event order are needed to evaluate multi-event models.
- **When a neural model pays off:** Cox is optimal for linear signal and collapses for non-linear signal; with few patients ($\leq 2\,500$) Cox and the transformer are comparable, from $\approx 5\,000$ patients the transformer leads clearly.
- **Open:** coded sequences (§4.2) do not yet beat a bag-of-codes table, the end-to-end LLM (§3) works but is clearly weaker, and continuous vs. discretised input is dataset-dependent.

1 Setup

1.1 Datasets

Dataset	n	events	regime	features
METABRIC	1 904	1	single event	9 (5 numeric, 4 categorical)
SUPPORT	8 873	1	single event	14 (8 numeric, 6 categorical)
hsa_synthetic	5 000	2	non-competing	15 numeric
EBMT	2 279	5	semi-competing (death censors others)	4 categorical
sim_* (simulator)	500–10 000	2–8	any	15 numeric, true CIFs known
ehrsim / ehrrsim2	5 000	4	semi-competing	coded visit histories (age, sex + ≈ 70 –80 codes)

The simulator (`sat/data/simulate.py`) draws Weibull proportional-hazard event times with controllable event-order spread, shared risk, frailty, non-linearity, censoring and regime; its true cumulative incidence functions (CIFs) give an *oracle* row. The EHR simulator (`sat/data/simulate_ehr.py`) generates diagnosis, medication, lab and noise codes over 5 years of visits; v2 removes shortcuts so that part of the risk is only visible in the order, trend and recency of the history.

1.2 Models

- **Ours:** numeric-value-embedding transformer + piecewise-constant-hazard head. Loss variants: $\mathcal{L}_{\text{PCH}}$; $+\mathcal{L}_{\text{rank}}$ (across patients, Eq. 4); $+\mathcal{L}_{\text{mul}}$ (within patient, Eq. 5); the paper recipe $\mathcal{L}_{\text{PCH}}+\mathcal{L}_{\text{rank}}+\mathcal{L}_{\text{mul}}$; and $+\text{MMV}$ (UniSurv mean/variance loss, weight λ_v).
- **Baselines:** cause-specific Cox PH; DeepHit, DSM and MENSA as published (same transformer backbone, their own heads/losses); the oracle (true CIFs) on simulated data.
- **End-to-end LLM** (§3): DistilGPT2 (pretrained) or a small GPT-2 (from scratch) answering with one token per time unit (N, C, Ek); curves derived from token probabilities.

1.3 Metrics

$C_{td} \uparrow$	Time-dependent concordance across patients, per event, at the 25/50/75 % quantiles (SurvTRACE protocol; multi-event: each event at its own quantiles), averaged.
Brier $\downarrow$	Calibration at the same horizons.
Within-subject C $\uparrow$	Eq. 5 pairs: at the time an event happened, was it ranked more likely than the patient’s later or censored events? <i>Can be gamed</i> (Section 4).
True-order C / Brier / log-loss	Simulated data only: predicted $P(T_a < T_b \mid x)$ from each model’s curves scored against the true (latent) event times; Brier and log-loss are proper scores (the oracle cannot be beaten in expectation).
MAE (uncensored / margin)	Error of the predicted event time in the dataset’s unit (METABRIC: months; SUPPORT, EBMT: days; simulator: arbitrary); margin also scores censored patients via a Kaplan–Meier best guess (unreliable on EBMT).

1.4 Protocol

Each configuration is run with 3–5 seeds; seed s uses initialisation s and train/validation/test split s (60/10/30 %). Checkpoints are selected on validation C_{td} . Tables report mean $\pm$ sd over seeds; “paired” comparisons use the same seed (same split) for both models and give the number of wins and a paired t -test p -value. With 5 seeds the smallest attainable Wilcoxon p is 0.0625, so we quote the t -test.

Earlier results are superseded. Before these runs, five bugs were fixed: train/validation were swapped (models trained on $\approx 10\%$ of the data), Cox ignored categorical features, the “ $+\mathcal{L}_{\text{mul}}$ ” run never contained $\mathcal{L}_{\text{mul}}$, checkpoints were selected with a broken C-index, and the split never changed across seeds.

2 Single-event benchmarks: METABRIC and SUPPORT (notebook 03)

Table 1: C_{td} (mean over the 25/50/75 % horizons) and Brier, 5 splits. “published” = SurvTRACE (Wang & Sun, 2022) Table 2, averaged over the same three horizons (their 10 runs, different splits). MAE-unc. in months (METABRIC) and days (SUPPORT). SUPPORT columns are affected by the preprocessing issue noted on page 1 and will be replaced.

Model	METABRIC				SUPPORT			
	C_{td}	Brier	MAE-unc.	published	C_{td}	Brier	MAE-unc.	published
Cox PH	0.625 ± 0.027	0.180	62.2	0.629	0.555 ± 0.004	0.203	608	0.566
DeepHit	0.670 ± 0.013	0.189	75.4	0.657	0.599 ± 0.005	0.218	545	0.609
DSM	0.649 ± 0.048	0.206	70.1	0.669	0.542 ± 0.042	0.253	578	0.615
MENSA	0.650 ± 0.027	0.188	69.2	–	0.581 ± 0.026	0.212	535	–
SurvTRACE (published only)	–	–	–	0.691	–	–	–	0.640
Ours $\mathcal{L}_{\text{PCH}}$	0.675 ± 0.014	0.177	69.6	0.679 [†]	0.608 ± 0.005	0.199	573	0.626 [†]
Ours $+\mathcal{L}_{\text{rank}}$	0.664 ± 0.008	0.185	66.8		0.608 ± 0.009	0.204	585	
Ours $+\text{MMV}$ ($\lambda_v=0.01$)	0.666 ± 0.011	0.190	59.0		0.593 ± 0.026	0.202	557	
Ours full, $\lambda_v=0.01$	0.658 ± 0.015	0.196	57.4		0.608 ± 0.007	0.204	577	
Ours full, $\lambda_v=0.1$	0.667 ± 0.012	0.189	56.6		0.607 ± 0.012	0.204	556	
Ours full, $\lambda_v=1$	0.669 ± 0.012	0.244	54.1		0.608 ± 0.004	0.205	494	

[†]published PC-Hazard (the same head without our transformer); “full” = $\mathcal{L}_{\text{PCH}}+\mathcal{L}_{\text{rank}}+\mathcal{L}_{\text{mul}}+\text{MMV}$.

Reading Table 1. Among the models we ran, our $\mathcal{L}_{\text{PCH}}$ model is the best on both datasets in discrimination and calibration. On METABRIC it matches the published PC-Hazard value (0.675 vs. 0.679) and Cox matches its published value (0.625 vs. 0.629), which validates the evaluation pipeline. On SUPPORT every model sits 0.01–0.02 below its published value, consistent with different splits. The published SurvTRACE remains ahead (0.691 and 0.640; 0.686 and 0.636 without its multi-task head): our backbone was tuned per dataset on validation over SurvTRACE’s own grid (METABRIC: 2 layers, SUPPORT: 3 layers; hidden size 16, lr 10^{-3}), but our model has no multi-task head and is evaluated on different splits, so parity with SurvTRACE is not shown. DSM is unstable (sd up to 0.048). Adding $\mathcal{L}_{\text{rank}}$ never helps (METABRIC -0.010 , 0/5 wins; SUPPORT ± 0) and worsens Brier in 5/5 splits on both datasets. MMV does what it targets: the error on observed patients falls from 69.6 to 54.1 months (METABRIC) and 573 to 494 days (SUPPORT) at $\lambda_v = 1$, but on METABRIC Brier rises from 0.177 to 0.244. On SUPPORT, $\lambda_v = 1$ is the best trade-off (same C_{td} , -14% MAE, $+0.006$ Brier).

3 Multi-event benchmarks (notebook 01, reruns in 07)

Table 2: Multi-event datasets, 5 splits (EBMT: 3–5, see text). Within-C = within-subject C-index (gameable); KM = the same score from population Kaplan–Meier curves (no patient information).

Model	sim_base (4 events)			hsa (2 events)	EBMT (5 events)	
	C_{td}	Brier	Within-C	C_{td}	C_{td}	Within-C
Oracle	0.736	0.156	0.696	–	–	–
Population KM	–	–	0.580	–	–	0.837
Cox PH	0.679	0.175	0.647	0.492	0.533	0.843
DeepHit	0.712	0.230	0.678	0.865	0.528	0.841
DSM	0.712	0.167	0.668	0.865	0.501	0.189
MENSA	0.701	0.172	0.649	0.865	0.504	0.465
Ours $\mathcal{L}_{PCH}$	0.713	0.166	0.674	0.867	0.545	0.840
Ours $+\mathcal{L}_{mul}$	0.700	0.172	0.666	0.846	0.535	0.880
Ours $+\mathcal{L}_{rank}$	0.713	0.180	0.639	0.868	0.543	0.838
Ours paper recipe	0.713	0.182	0.658	0.864	0.537	0.879
Ours full (+MMV, $\lambda_v=1$)	0.712	0.213	0.673	0.858	0.550	0.894

Reading Table 2. On the simulator our $\mathcal{L}_{PCH}$ model is the closest learned model to the oracle (C_{td} 0.713 vs. 0.736) with the best Brier; DeepHit and DSM tie on C_{td} but are worse calibrated. On hsa_synthetic the signal is purely non-linear: Cox is at chance (0.49) while all neural models reach ≈ 0.865 . On EBMT the four categorical covariates carry little signal ($C_{td} \approx 0.53$ – 0.55 for every model) and the population order of events is already very predictable (KM within-C 0.837). Adding $\mathcal{L}_{mul}$ lowers C_{td} on sim_base (-0.013 , 0/5 wins, $p = 0.002$) and hsa (-0.021 , 0/5, $p < 0.001$). On EBMT it raises within-C from 0.84 to 0.88 (5/5 wins) — but EBMT is semi-competing, exactly the setting where Section 4 shows this metric is gamed. The three EBMT runs that crashed (fixed, notebook 07) agree with the table ($\mathcal{L}_{PCH}+\mathcal{L}_{mul}$: C_{td} 0.537, within-C 0.887; full: 0.525, 0.901).

4 Is within-subject ordering really improved? (notebook 08)

Table 3: Within-subject ordering scored against the simulator’s *true* event times, 5 splits. True-order Brier and log-loss are proper scores: no model can beat the oracle in expectation.

Model	sim_base (non-competing)					semi-competing			
	C_{td}	Within-C	TO-C	TO-Brier	TO-logloss	C_{td}	Within-C	TO-C	TO-Brier
Oracle	0.736	0.696	0.685	0.203	0.592	0.734	0.739	0.673	0.210
Cox PH	0.679	0.647	0.624	0.227	0.645	0.679	0.716	0.623	0.227
DeepHit	0.711	0.674	0.552	0.244	0.681	0.712	0.546	0.559	0.281
MENSA	0.709	0.664	0.653	0.218	0.626	0.708	0.715	0.644	0.221
Ours $\mathcal{L}_{PCH}$	0.714	0.673	0.659	0.215	0.618	0.714	0.738	0.658	0.214
Ours $+\mathcal{L}_{mul}$	0.709	0.671	0.657	0.220	0.630	0.703	0.787	0.633	0.228
Ours paper recipe	0.715	0.659	0.653	0.230	0.653	0.713	0.781	0.618	0.232

TO = true order. Red: above the oracle on the gameable metric while worse on every true-order score.

Reading Table 3. This is the decisive test for $\mathcal{L}_{mul}$. On true-order scores our $\mathcal{L}_{PCH}$ model is the best learned model in both regimes, ahead of MENSA, Cox and DeepHit. Adding $\mathcal{L}_{mul}$ or using the paper recipe makes every true-order score worse (TO-Brier: 0/5 wins in all four comparisons, $p < 0.001$). In the semi-competing regime $\mathcal{L}_{mul}$ raises the gameable within-C to 0.787 — *above the oracle’s* 0.739 — while its true-order C falls by 0.025. The loss is trained on the same pairs the metric counts and learns to reshape the event curves around the terminal event, which wins those

pairs without predicting better. DeepHit has a good C_{td} but the worst true ordering (0.552): its likelihood assumes one event per patient (competing risks) and is misspecified when events co-occur.

Tuning the penalties

Table 4: Penalty weight and sharpness σ tuned on validation C_{td} (split 0), then tested on 5 splits.

Penalty (data)	chosen setting		C_{td}	Brier	TO-Brier
$\mathcal{L}_{\text{rank}}$ (METABRIC)	$\sigma=0.5$, weight 0.01	$\mathcal{L}_{\text{PCH}}$	0.6705	0.1810	—
		tuned $\mathcal{L}_{\text{rank}}$	0.6718	0.1776	—
$\mathcal{L}_{\text{mul}}$ (sim_base)	$\sigma=0.5$, weight 0.2	$\mathcal{L}_{\text{PCH}}$	0.7152	0.1662	0.2148
		tuned $\mathcal{L}_{\text{mul}}$	0.7139	0.1661	0.2145

Reading Table 4. The validation grids ($\sigma \in \{0.1, 0.5, 1\}$, $\text{weight} \in \{0.01, 0.05, 0.2\}$) both ranked $\sigma = 0.1$ — the value used everywhere before — last. With the best setting each penalty becomes indistinguishable from $\mathcal{L}_{\text{PCH}}$ (METABRIC 3/5 wins, $p = 0.81$). Tuning removes the harm but adds no benefit. Our reading: $\mathcal{L}_{\text{PCH}}$ is the full likelihood of the event times, so it already learns the ranking and the order the model can represent; a ranking penalty can only pull the curves away from that optimum.

5 Simulator sweeps (notebooks 02 and 11)

Each figure varies one simulator setting around the base scenario (3 splits; error bars = sd). The dashed grey line is the oracle. Panels: within-subject C, C_{td} , within-C gain over population KM (and, for notebook 11, true-order Brier). Notebook 02 ran before the true-order score existed.

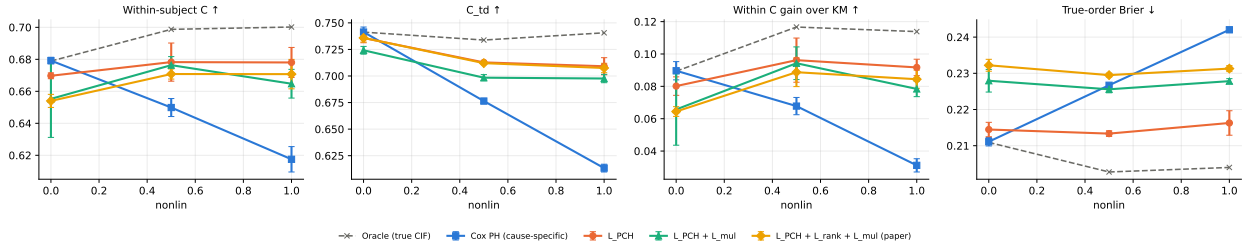


Figure 1: Non-linearity of the risk (0 = linear, 1 = fully non-linear).

Figure 1. With a linear signal Cox equals the oracle (C_{td} 0.741) and our model loses little (0.736). As the signal becomes non-linear Cox collapses (0.613) while ours stays at 0.709 (oracle 0.741). This is the clearest argument for a neural model.

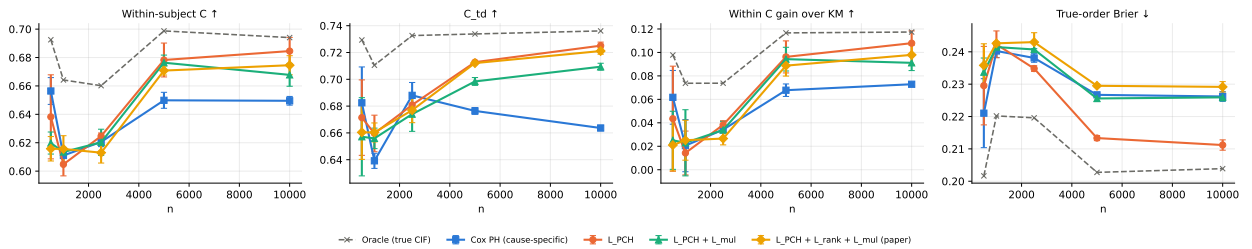


Figure 2: Training-set size (500 to 10000 subjects).

Figure 2. The gap of our model to the oracle shrinks from 0.058 ($n=500$) to 0.011 ($n=10000$); Cox’s gap stays 0.05–0.07. Below 5000 subjects the two are comparable (Cox ahead at $n = 500$ and

2500, the transformer at $n = 1000$); from ≈ 5000 subjects the transformer leads clearly. Each n is a different random dataset, so the small- n points are noisy.

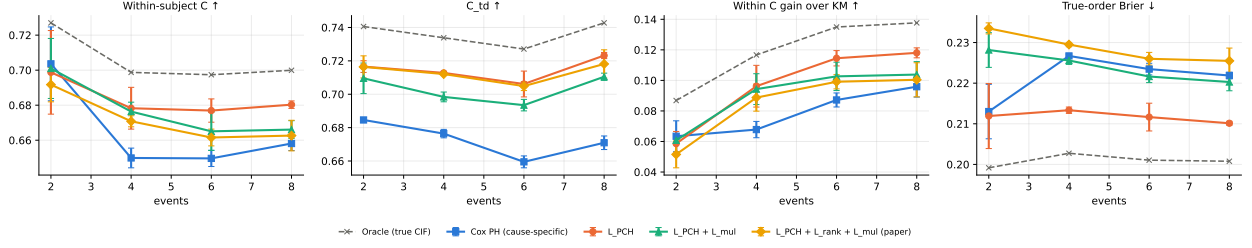


Figure 3: Number of events K (2 to 8).

Figure 3. Our model stays ≈ 0.02 below the oracle for every K while Cox stays 0.05–0.07 below: the framework scales with the number of events. $\mathcal{L}_{mul}$ and the paper recipe are below $\mathcal{L}_{PCH}$ on true-order Brier at every K (e.g. 0.226 / 0.230 vs. 0.213 at $K=4$).

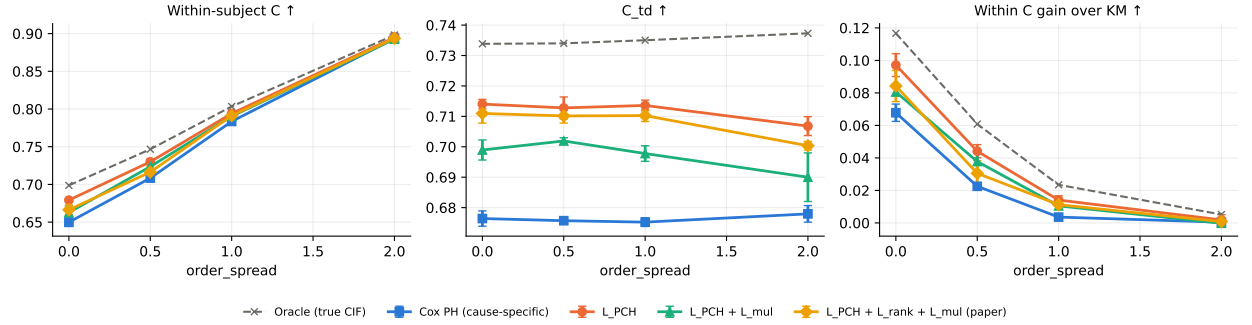


Figure 4: Spread of the events’ typical timing (0 = the patient’s covariates decide the order, 2 = nearly the same order for everybody).

Figure 4. When the order is population-wide (spread 2) every model reaches the oracle’s within- C (≈ 0.89 – 0.90) and there is nothing patient-specific to learn (gain ≈ 0). When the order is patient-specific (spread 0) the oracle reaches 0.699, $\mathcal{L}_{PCH}$ 0.679 and $\mathcal{L}_{mul}$ 0.663 — $\mathcal{L}_{mul}$ is lower exactly where it was meant to help.

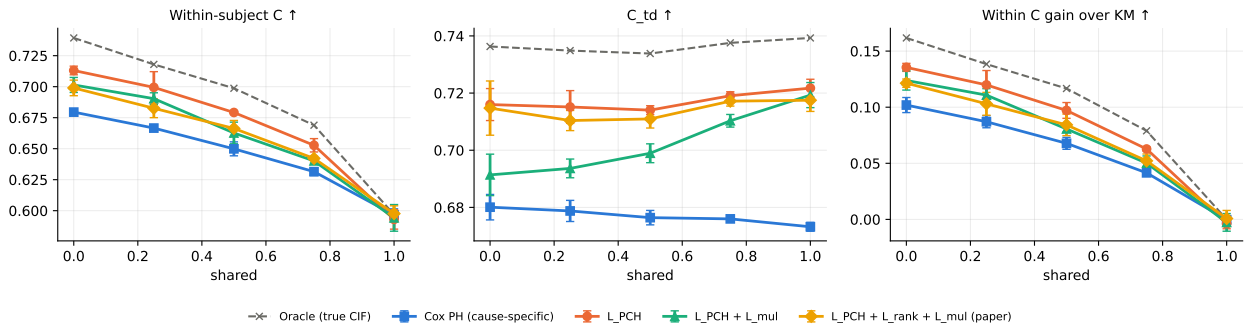


Figure 5: Share of the risk common to all events (“shared disease pathway”).

Figure 5. With more shared risk there is less patient-specific ordering to learn; at shared = 1 even the oracle gains nothing over population ordering (0.000), confirming the metric behaves as intended in the non-competing regime. Our model tracks the oracle at every level; $\mathcal{L}_{mul}$ is never above $\mathcal{L}_{PCH}$.

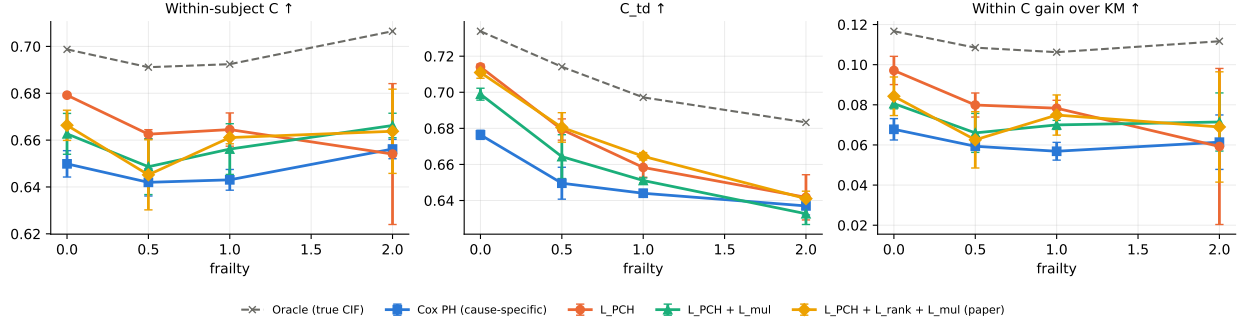


Figure 6: Unobserved shared frailty (dependence covariates cannot explain).

Figure 6. Frailty lowers every model and the oracle together (C_{td} of $\mathcal{L}_{PCH}$ 0.714 $\rightarrow$ 0.642 at variance 2); no loss variant is more robust than $\mathcal{L}_{PCH}$.

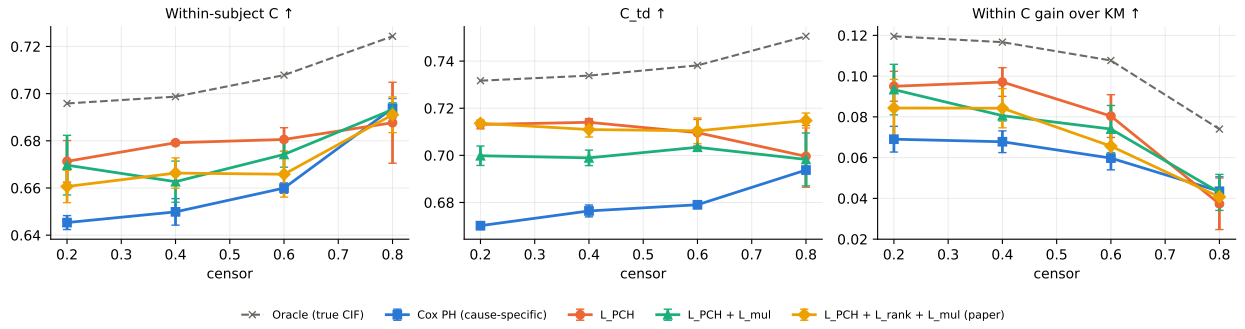


Figure 7: Fraction of unobserved event slots (censoring).

Figure 7. Performance is stable from 20 % to 80 % censoring ($\mathcal{L}_{PCH}$ C_{td} 0.700–0.714); the ordering of models does not change.

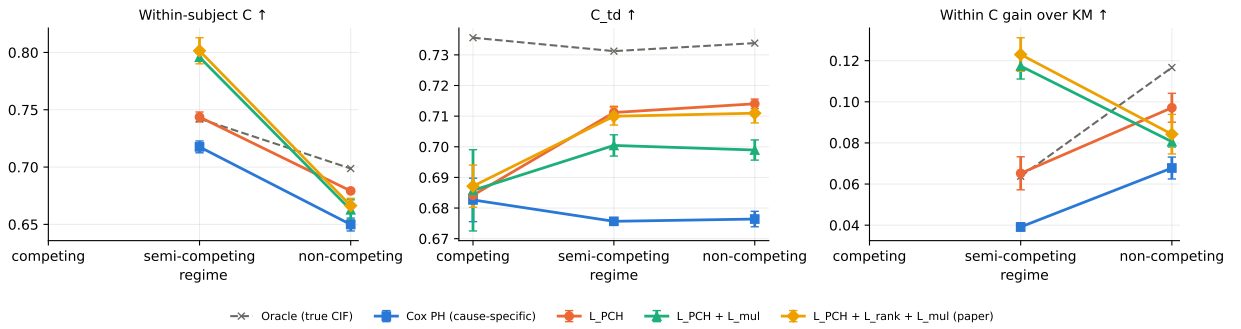


Figure 8: Observation regime.

Figure 8. Under competing risks only the first event is observed, so there are no orderable pairs and within-C is undefined. Under semi-competing risks $\mathcal{L}_{mul}$ and the paper recipe reach within-C 0.796–0.802, far above the oracle’s 0.742: the first sign of the gaming confirmed in Table 3. C_{td} is similar across regimes (≈ 0.70 –0.71 for our model).

6 Continuous vs. discretised numeric input (notebook 10)

Table 5: Same model and splits, numeric features standardised (continuous) or cut into 10 quantile bins. C_{td} , 5 splits. SUPPORT “continuous” is affected by the preprocessing issue noted on page 1 (the discretised path is not).

Model	METABRIC		SUPPORT	
	continuous	discretised	continuous	discretised
Ours $\mathcal{L}_{PCH}$	0.675 $\pm$ 0.011	0.652 $\pm$ 0.018	0.609 $\pm$ 0.009	0.621 $\pm$ 0.010
DeepHit	0.675 $\pm$ 0.010	0.654 $\pm$ 0.019	0.603 $\pm$ 0.009	0.625 $\pm$ 0.005

Reading Table 5. Continuous input is better on METABRIC (+0.023) and discretised input is better on SUPPORT (+0.012 to +0.022), for both models. Part or all of the SUPPORT difference is explained by the preprocessing issue noted on page 1: on the continuous path five SUPPORT measurements were unscaled and one was lost, while the discretised path was correct. The corrected comparison, including a quantile (rank) transform, is in notebook 12. The paper should not claim continuous encoding is always better; a rank/quantile transform before the value embedding is a cheap candidate to get both.

7 Coded visit histories, §4.2 (notebooks 06 and 09)

Table 6: The same patients given to our model as the full time-stamped sequence, a bag-of-codes table (counts, last lab values, number of visits) or static features only. C_{td} , 3 splits. “temporal” = share of risk carried by order, trend and recency.

Design	cohort	Oracle	Cox [bag]	Ours [bag]	Ours [sequence]	MENSA [seq.]	Ours [static]
v1	static (0.0)	0.735	0.725	0.710	0.708	0.712	0.502
v1	base (0.5)	0.723	0.708	0.705	0.700	0.690	0.500
v1	temporal (0.9)	0.708	0.688	0.686	0.684	0.675	0.500
v2	level (0.0)	0.745	0.735	0.716	0.726	0.721	0.502
v2	mixed (0.5)	0.725	0.697	0.691	0.687	0.681	0.498
v2	order (0.9)	0.696	0.653	0.655	0.647	0.645	0.502

Reading Table 6. In v1 the temporal signal could be recovered from counts and last values (a linear model on the bag recovers 66–86 % of the true risk), so sequence and bag tie. v2 removes this shortcut (the bag recovers only 12–41 % for three of four outcomes), and the oracle’s lead over every model grows with the temporal share (0.010 $\rightarrow$ 0.043): the extra information is there. No model extracts it: our sequence model stays at the bag level (0.647 vs. 0.655 in the “order” cohort). Likely causes are the very small transformer (hidden size 16, 2 layers), mean pooling over up to 256 mostly-noise tokens, time entering only as the value of a VISIT token, and $\approx$ 3 000 training patients. Static input (age and sex) is at chance, as designed. §4.2 is therefore not supported yet. On these data DeepHit again orders events badly (within-C 0.41–0.47).

8 End-to-end LLM, §3 (notebooks 05 and 07)

Table 7: The §3 language model vs. Cox and our transformer on the same splits. C_{td} , 3 splits.

Model	sim_base	EBMT	METABRIC
Oracle	0.734	–	–
Cox PH	0.676	0.538	0.624
Ours $\mathcal{L}_{PCH}$	0.713	0.544	0.666
Ours paper recipe	0.709	0.544	0.676
LLM, DistilGPT2 pretrained	0.621 ± 0.060	0.550	0.589 ± 0.038
LLM, GPT-2 from scratch	0.654 ± 0.012	0.538	0.544 ± 0.055

Reading Table 7. The LLM formulation works end to end and is clearly above chance, but it is weaker than our transformer by 0.06–0.08 on sim_base and METABRIC; on EBMT, where every model is near 0.54, it ties. The pretrained model is unstable across splits (sd up to 0.06). It is best presented as a proof of concept.

9 What we know now

1. The transformer with the $\mathcal{L}_{PCH}$ likelihood is the best or tied-best of the models we ran on METABRIC, SUPPORT, hsa_synthetic, EBMT and the simulator, with the best calibration among neural models (Tables 1, 2); it is below the published SurvTRACE on METABRIC and SUPPORT. On the synthetic EHR cohorts, whose risk is close to linear in the bag-of-codes features, Cox on the bag table is equal or better (Table 6).
2. Measured against the true event order, it orders events within a patient better than every other learned model; the competing-risk DeepHit is worst (Table 3). This supports the general multi-event formulation.
3. $\mathcal{L}_{rank}$ and $\mathcal{L}_{mul}$ do not improve any proper score, tuned or not (Tables 3, 4; Figures 3 and 4).
4. The within-subject C-index rewards a loss trained on it beyond the oracle (Table 3, Figure 8); proper true-order scores are needed to evaluate multi-event models.
5. The neural model pays off for non-linear signal and clearly from $\approx 5\,000$ subjects; it scales to 8 events (Figures 1–3).
6. MMV trades calibration for lower error on observed event times; useful only when time accuracy matters.
7. Continuous vs. discretised input is dataset-dependent (Table 5).
8. Coded sequences do not yet beat a bag of codes although the signal is present (Table 6).
9. The end-to-end LLM works but is weaker (Table 7).

10 Open items

Ordered by how much each could change the paper’s claims.

1. **Corrected SUPPORT results.** Re-run of the SUPPORT benchmark and of the encoding comparison with the fixed preprocessing (notebook 12); this also re-opens the gap to SurvTRACE on SUPPORT.
2. **Coded sequences (§4.2).** The v2 cohorts show the information is in the history but our model does not extract it. Planned: a larger transformer, attention or [CLS] pooling instead of mean pooling, an explicit time encoding, and more training patients; then re-run notebook 09. This is the one negative result that could still turn into support for the paper.
3. **Gap to the published SurvTRACE.** Our model is 0.016 (METABRIC) and 0.032 (SUPPORT) below it. The backbone is already tuned per dataset; what remains is SurvTRACE’s multi-task head (worth 0.004–0.005 in their own ablation) and a comparison on identical splits before claiming parity.

4. **Numeric encoding.** Continuous input wins on METABRIC and quantile bins win on SUPPORT; a rank/quantile transform before the value embedding may give both.
5. **Role of $\mathcal{L}_{\text{rank}}/\mathcal{L}_{\text{mul}}$ in the paper.** The evidence says they are neutral when tuned and harmful when sharp. The paper should either present them as optional penalties with this ablation, or drop the claim that they improve prediction.
6. **Evaluation of multi-event models.** The within-subject C-index is gameable; the true-order scores (Section 4) should replace it in the paper, which is also a methodological contribution.
7. **Real multi-event data with richer covariates.** EBMT has only four categorical covariates. MIMIC needs PhysioNet credentialing; the §4.2 CKD/COPD/diabetes cohorts can be loaded through `patients.csv` and `events.csv` once available.
8. **Manuscript corrections.** The sign of η in Eqs. 4–5 (the code is correct), a MENSA row in Table 1 of the paper, and the end-to-end LLM presented as a proof of concept.

A Where the numbers come from

Table / figure	source (results/new_results/)
Table 1	03_single_event_benchmarks/report
Table 2	01_multievent_lmml/report, 07_rerun_failed/report
Tables 3, 4	08_true_order_and_tuning (tuned and untuned runs separated by tag)
Figures 4–8	02_synthetic_scenarios/report/sweep_*
Figures 1–3	11_extra_sweeps/report/sweep_*
Table 5	10_continuous_vs_discretised/report
Table 6	06_sequential_ehr/report, 09_sequential_ehr_v2/report
Table 7	05_llm_end_to_end/report, 07_rerun_failed/report